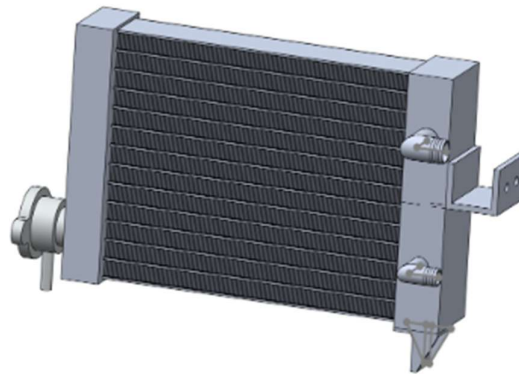




DESIGN REPORT:

Radiator Thermal Design



ϵ -NTU Modelling & Transient Validation for the EV Powertrain Cooling System

Darsh Raghuvanshi

Drivetrain Subdivision

Junior Design Engineer - IIT Bombay Racing



Contents

1. Introduction and Overview	3
1.1 GitHub Repository.....	3
1.2 Physics Background: The ϵ -NTU Method.....	4
1.3 Notation (Kays & London Convention)	4
2. Steady-State Design Script (radiator_steady.mlx).....	5
2.1 Required Heat Rejection.....	5
2.2 Coolant (Water) Properties via CoolProp	5
2.3 Scrapped Methodology	6
2.4 Fixed Geometry & Sweep Variables	7
2.5 Water-Side Convection Coefficient.....	7
2.6 Air-Side Convection Coefficient	7
2.7 Overall Conductance and Capacity Rates.....	8
2.8 Fin Efficiency / Effective Area and Fin Effectiveness	8
2.9 NTU, ϵ , and Predicted Heat Rejection.....	10
2.10 Extracting the Design Point.....	10
2.11 Water-Side Pressure Drop	11
2.12 Sensitivity Studies	12
2.13 A_face vs. FPI Summary Table.....	13
3. Transient Models (radiator_transient_fmot.mlx / _emrax.mlx)	14
3.1 Extracting Data as Input	14
3.2 Data Cleaning: Zero-RPM Filtering & Time Shifting.....	14
3.3 Motor + Controller Efficiency Map.....	15
3.4 Initial Parameters (Flagged for Iteration)	16
3.5 Deriving Air Velocity from Wheel Kinematics.....	16
3.6 Q_epsilonNTU Function	16
3.7 Time-Marching Loop (Lumped Capacitance, Explicit Euler)	17
3.8 Outputs	17
4. F-MOT vs. EMRAX: What's Different	18
5. Validation Suite (radiator_transient_test.mlx)	18
5.1 Independent Steady-State Solver.....	18
5.2 Test Case Matrix.....	19
5.3 Time Constant and t99.....	19
5.4 Summary Table and Pass/Fail Checks.....	20
6. COMSOL Validation of the Steady-State Model.....	20
6.1 Full Air and Water Domain Simulations (Iterations 1–5)	21
6.2 Reduced Geometry and Modelling Flaw (Iterations 6–7)	23
6.3 Periodic Unit Cell Model (Iterations 8–10)	25
6.4 Air Temperature as a Boundary Condition (Iteration 11).....	27
6.5 Implementation Details.....	28
6.6 Results and Comparison with the MATLAB Model	28
7. Known Assumptions and Simplifications	29
8. Suggested Workflow	30

File Contents

This report documents the thermal sizing and validation of the radiator used to cool the 2WD powertrain (one motor and one motor controller per side). The work is split into a steady-state design/sizing tool and a transient (time-marching) validation tool, run against real endurance-run data for the outgoing motor, plus a synthetic test suite that validates the transient solver before its tested with real test-run data.

File	Purpose
<code>radiator_steady_final.mlx</code>	Core sizing tool. Sweeps FPI and frontal area to find the minimum radiator geometry that rejects the target heat load, using the ϵ -NTU method.
<code>radiator_transient_fmot.mlx</code>	Calculates coolant temperature over a real endurance-run data set for the F-MOT motor, using the geometry chosen in the steady script.
<code>radiator_transient_emrax.mlx</code>	Identical structure to the F-MOT script, but with the EMRAX motor + controller efficiency map substituted in.
<code>radiator_transient_test.mlx</code>	Constant-condition test cases that validate the transient model against an independently solved steady-state temperature, before the model is trusted against actual data.

Each transient script has its own copy of the `Q_epsilonNTU` function, the ϵ -NTU heat-rejection calculation (identical to the steady-state script), so it can be called once per timestep.

1. Introduction and Overview

This file documents the methodology, assumptions, and current state of the modelling toolchain, and is intended to be read alongside the MATLAB scripts.

The toolchain is split into two complementary tools:

- A **steady-state design / sizing tool** (`radiator_steady_final.mlx`) that sweeps fin density and frontal area to find the minimum radiator geometry that clears a fixed, conservative worst-case heat load.
- A **transient (time-marching) validation tool** (`radiator_transient_fmot.mlx` / `radiator_transient_emrax.mlx`) that replays real endurance-run data through the chosen geometry to optimize the design (as steady state assumes worst-case conditions, radiator will be over optimized initially) for the actual accel / brake / idle profile of an event.

A **synthetic validation suite** (`radiator_transient_test.mlx`) tests the transient solver against constant-condition cases with an independently-solved steady-state answer, so the transient solver implementation is trusted before it is run against real data.

1.1 GitHub Repository

This document is meant to be read along with all the codes which can be found in the attached GitHub Repository. Coding intricacies like passing constants as structs to functions

are not included in this report so would recommend going through both the report and the codes in length.

https://github.com/darshsys/Radiator_design

1.2 Physics Background: The ϵ -NTU Method

Sizing a radiator means finding the heat rejected, Q , for a given geometry and a given set of operating conditions, without needing to iteratively guess an outlet temperature. In the steady state effectively all temperatures are known or assumed ($T_{coolant_in}$ fixed, $T_{coolant_out}$ capped due to limit on motor controller, T_{air} is ambient temp and T_{air_out} can be calculated via energy balance on Q).

This selection only becomes apparent once the model moves to the transient state. At every timestep, Q is taken as an input and the updated $T_{coolant_out}$ is calculated. LMTD is problematic in this formulation because it requires the outlet temperature to calculate the log-mean temperature difference, creating a circular dependency. ϵ -NTU avoids this because ϵ is a function of NTU and C_r alone, both computable from known quantities (UA from geometry and C_{min}/C_{max} from mass flow rate and C_p) without prior knowledge of the outlet temperature. Because the transient model requires ϵ -NTU on these grounds, the steady-state sizing model was also implemented in ϵ -NTU.

- **Effectiveness, ϵ :** the ratio of actual heat transfer to the maximum thermodynamically possible heat transfer (i.e. if one fluid stream were brought all the way to the other stream's inlet temperature).
- **NTU (Number of Transfer Units):** $NTU = UA / C_{min}$, a non-dimensional measure of how large the heat exchanger is relative to the smaller of the two fluid capacity rates.
- **C_{min} , C_{max} , C^* :** the fluid capacity rates $C = \dot{m} \cdot c_p$ for each stream; C_{min} is the smaller of the two (air or water side), $C^* = C_{min} / C_{max}$.

Given ϵ , $Q = \epsilon \cdot C_{min} \cdot (T_{coolant_in} - T_{air_in})$, with no need to solve for the outlet temperatures directly.

All scripts use the same closed-form correlation for a **crossflow heat exchanger with both fluids unmixed** (the standard configuration for a fin-and-tube radiator), taken from Kays & London:

```
epsilon = 1 - exp((NTU^0.22/C_star)*(exp(-C_star*NTU^0.78)-1));
```

Everything else in the scripts exists to compute the two inputs this formula needs: $U \cdot A$ (the overall thermal conductance) and C_{min}/C^* .

1.3 Notation (Kays & London Convention)

The following symbol convention is used throughout all four scripts and this report:

Symbol	Meaning
A_{fr} / A_{face}	Frontal (face) area of the radiator core

Symbol	Meaning
A_o / A_{fins}	Effective, fin-efficiency-corrected air-side heat transfer area
σ (σ_c)	Contraction ratio: ratio of minimum free-flow area to frontal area
G	Mass flux through the minimum free-flow area ($\dot{m}/A_{min}$)
D_h	Hydraulic diameter of a flow passage
j	Colburn j-factor (heat transfer correlation)
f	Friction factor (pressure drop correlation)
NTU, ϵ , C_{min} , C_{max} , C^*	Standard heat-exchanger effectiveness terms

2. Steady-State Design Script (radiator_steady_final.mlx)

This is the primary sizing tool: for a grid of candidate frontal areas and fin densities, it computes predicted heat rejection and identifies the minimum geometry that clears the required heat load.

2.1 Required Heat Rejection

The motor's peak power and its worst-case (lowest) efficiency point from the datasheet are used to get a single conservative heat-loss number. The motor controller's dissipation is taken as a fixed worst-case figure rather than a function of load.

```
import py.CoolProp.CoolProp.*
P_peak = 35000; % peak operating power of motor (W), F-MOT motor datasheet
efficiency = 0.93; % worst efficiency according to datasheet (%)
Q_motor = P_peak - efficiency*P_peak; % heat loss through water (W)
Q_motor_controller = 750; % max power dissipation according to datasheet, (W)
Q_target = Q_motor + Q_motor_controller; % (W)
```

Flag: this fixed 750 W MC figure and the transient model's map-based combined motor+MC loss are two different accounting methods for the same physical dissipation.

2.2 Coolant (Water) Properties via CoolProp

Water properties are pulled from CoolProp (PropsSI, via MATLAB's Python interop) rather than hardcoded, so they can be re-evaluated at the correct bulk temperature. Density and specific heat are pulled once at a fixed reference temperature (50°C) since they barely change over the coolant's operating range. Once Q_{target} and $\dot{m}_{coolant}$ give a first-pass ΔT and average bulk temperature, viscosity, conductivity, and Prandtl number are re-evaluated at that average temperature, since these three properties matter more for the Reynolds/Nusselt calculation and are more temperature-sensitive.

Thermal conductivity (k_{water}), Prandtl number (Pr_{water}) and dynamic viscosity

(μ_{water}) were chosen to be recalculated after temperature is fixed because they vary substantially within the operating temperature range (20°C-70°C):

Water Properties (20°C vs 70°C)			
Property	20°C	70°C	% Change
Thermal Cond. (k) [W/m-K]	0.598	0.663	+10.9%
Dyn. Viscosity (μ) [mPa-s]	1.002	0.404	-59.7%
Prandtl No. (Pr)	7.01	2.55	-63.6%

Figure 1: Variation of water properties across temperature range of 20°C-70°C

```
rho_water = PropsSI('D', 'T', 323.15, 'P', P_water, 'Water');
cp_water = PropsSI('C', 'T', 323.15, 'P', P_water, 'Water'); % using 323.15K

k_water = PropsSI('L', 'T', T_avg_K, 'P', P_water, 'Water'); % using T_avg_K
mu_water = PropsSI('V', 'T', T_avg_K, 'P', P_water, 'Water');
Pr_water = PropsSI('Prandtl', 'T', T_avg_K, 'P', P_water, 'Water');
```

Flag: rather than assuming 50°C, a more accurate bulk temperature value can be taken once the transient model is run on simulated data of the new motors (F-MOT).

2.3 Scrapped Methodology

On the first iteration of the code, rather than using f_{pi} as the sweep variable, fin density (σ , m^2/m^3) was swept directly as a measure of surface area added per unit volume occupied by the fins. This method was later scrapped in favour of computing fin efficiency, which needed concrete geometric numbers (fin height, fin thickness) rather than an arbitrary density value, and so that realistic numbers could be given to the manufacturer to get their opinion on manufacturing feasibility. Accordingly, this code snippet was commented out and not used in the final script:

```
% function [FPI] = sigma_to_fpi(sigma,f_height,f_thickness)
% p = 2*(f_height-f_thickness)/(sigma*f_height-2); %fin pitch (m)
% p_in = p/0.0254; %pitch (inches)
% FPI = 1/p_in;
% end
```

2.4 Fixed Geometry & Sweep Variables

Fin height (8 mm) and fin thickness (0.1 mm) are carried over from the senior team's reference script and held fixed - they are not being re-optimized in this iteration. The two free design variables being swept are `fpi_range` (fins per inch, 8 to 25) and `A_face_range` (radiator frontal area, 0.01 to 0.20 m²). Radiator height `H` is fixed at 0.2 m for the main sweep (its sensitivity is checked separately).

Flag: in the later phases of manufacturing, more accurate fin height and fin thickness numbers can be taken after talking with the manufacturer.

2.5 Water-Side Convection Coefficient (`h_water`)

The coolant tube is treated as a rectangular duct of thickness 0.3mm and outer dimensions as 2mm by 32mm (d). For dimensions of water flow channel 2*thickness is subtracted from outer tube dimensions. Its aspect ratio, hydraulic diameter and cross-sectional area set up a standard internal-flow correlation:

```
a_tube = 0.002; % dimensions of rectangular tube, height as 2mm
b_tube = d; % width of tube taken equal to depth of radiator
a_water = a_tube-0.0003*2; % tube dimension = 0.3mm
b_water = b_tube-0.0003*2; % these are dimensions of water channel
alpha = a_water/b_water; % aspect ratio

tube_pitch = a_tube + f_height; % height of tube(2mm) + fin space(8mm), used to
calculate number of tubes
D_h_tube = (4*a_water*b_water)/(2*(a_water+b_water)); % hydraulic diameter of tube
A_tube = a_water*b_water; % tube cross sectional area (m^2)
```

Flow regime is checked explicitly: laminar flow ($Re < 2300$). Tube length required for temperature profile to fully develop is given by $L_{th} \approx 0.05*Re*Pr*D_h$ which for our case comes out to be approximately 1 metre. This means out laminar flow doesn't fully develop in the water tube channel.

Nusselt number taken is from a curve fit on tabulated data in laminar flow in rectangular duct with aspect ratio taken as `alpha` from Shah & London and then a correction factor is applied to take into account the non-fully developed part of the flow.

```
n_tubes = floor(H/tube_pitch); %number of tubes
mdot_tube = mdot_coolant_kgs/(n_tubes); %mass flow rate per tube
v_water = mdot_tube/(rho_water*A_tube); %velocity of water in each tube
Re_water = (rho_water*v_water*D_h_tube)/mu_water;
if Re_water < 2300
    Nu_fd = 8.235*(1 - 2.0421*alpha + 3.0853*alpha^2 - 2.4765*alpha^3 +
1.0578*alpha^4 - 0.1861*alpha^5);
    % this is a curve fit from tabulated data from shah & London, flow is
    % assuming fully developed laminar flow in rectangular tube with aspect
    % ratio alpha
    Gz = (D_h_tube/W)*Re_water*Pr_water; % graetz number, takes it that flow
is developing across length of water channel
```

```

        Nu_water = Nu_fd + 0.0668*Gz/(1 + 0.04*Gz^(2/3)); % hausen entry
        correction, length-averaged Nu for thermally developing flow (enhances
        Nusselt number)
    else
        Nu_water = 0.023*(Re_water^0.8)*(Pr_water^0.3); %turbulent
    end
    h_water = (Nu_water*k_water)/D_h_tube;

```

Flag: Turbulent flow was not researched in as much depth as Laminar flow so the Nusselt number relation written is probably wrong. But since out flow is entirely in the laminar region its not an issue.

2.6 Air-Side Convection Coefficient (h_{air})

The air side is where most of the fin geometry detail lives:

- **Contraction ratio (σ):** as air enters the fin channels between tubes, the flow area contracts. σ_c captures this, and is used to convert the frontal-area-based approach velocity into the actual (higher) velocity through the minimum flow area.
- **Mass flux:** $G_{air} = \rho_{air} \cdot v_{air} / \sigma_c$. Captures the fact that when, for a constant mass flow rate, air travels faster through lesser area, so the frontal area disrupted by the tube and fins is taken into account using contraction ratio.
- **Fin channel hydraulic diameter:** from fin spacing ($0.0254/f_{pi} - f_{thickness}$) and fin height. 0.0254 is conversion factor from inches to metres.
- **Colburn j-factor:** $J_{air} = 0.174/(Re_{air})^{0.383}$, an empirical heat-transfer correlation for the fin geometry. This correlation is inherited from the prior code base and its original source/derivation has not yet been independently re-verified against the actual fin geometry class being used.
- **Colburn analogy:** $h_{air} = J_{air} \cdot G_{air} \cdot cp_{air} / Pr_{air}^{(2/3)}$ converts the j-factor into a convection coefficient.

```

%air side convection heat transfer coefficient
sigma_c = 1-((fpi*f_height*f_thickness)/0.0254+a_tube)/(a_tube+f_height);
%contraction factor
G_air = rho_air*v_air/sigma_c; % mass flux
f_spacing = 0.0254/fpi - f_thickness;
D_h_fin = (4*f_spacing*f_height)/(2*(f_spacing+f_height)); %hydraulic diameter of
fin channel
Re_air = G_air*D_h_fin/mu_air; %reynolds numbers
J_air = 0.174/(Re_air)^0.383; % Colburn j-factor
h_air = J_air*G_air*cp_air/(Pr_air^(2/3));

```

Flag: the Colburn j-factor value used here has not yet been independently confirmed against Kays & London's compact heat exchanger surface data for this fin geometry class.

2.7 Overall Conductance and Capacity Rates

This treats the tube wall as having negligible thermal resistance, and contact/bond resistance between fin and tube as negligible, a simplification pending manufacturing/bonding-method confirmation. Actual conductance factor is calculated later taking into consideration enhanced surface area by the fins and reduced surface area due to inclusion of tube thickness.

```
%capacity rates
C_air = rho_air*v_air*cp_air*A_face;
C_water = mdot_coolant_kgs*cp_water;
C_min = min(C_air,C_water);
C_max = max(C_air,C_water);
C_star = C_min/C_max;

% Thermal resistances
R_air = 1/(h_air*A_air_eff);
R_water = 1/(h_water*A_water_inner);

% Overall conductance
UA = 1/(R_air + R_water);
```

2.8 Fin Efficiency / Effective Area and Fin Effectiveness

There are 2 different heat transfer surfaces involved in the radiator: the bare tube wall in contact with air (A_{primary}) and the fin surface ($A_{\text{secondary}}$). Fins are not 100% efficient conductors, so a fin efficiency correction is applied.

Fin efficiency is evaluated using the standard analytical solution for a straight fin with an insulated tip approximation: $\eta_f = \frac{\tanh(mL_c)}{mL_c}$. This fraction represents the actual heat transfer from the fin divided by the ideal heat transfer if the entire fin remained at the root (tube wall) temperature.

Surface efficiency calculated as $\eta_0 = 1 - \frac{A_{\text{secondary}}}{A_{\text{primary}} + A_{\text{secondary}}} (1 - \eta_f)$. This creates a weighted average of the two distinct surfaces. The bare tube wall (A_{primary}) is treated as 100% efficient because it is in direct proximity to the coolant, while the fin surface ($A_{\text{secondary}}$) is penalized by its thermodynamic efficiency η_f .

```
%A_fins calculation
A_primary = 2*d*W*n_tubes*(1 - f_thickness*fpi/0.0254); % tube and air contact surface area
A_water_inner = 2*(a_water+b_water)*W*n_tubes; % tube and water contact area
A_secondary = 2*f_height*d*fpi*W*n_tubes/0.0254; % fin and air contact surface area
k_fin = 237; %thermal conductivity of aluminium (fin material)
% fin efficiency
m = sqrt(2*h_air/(k_fin*f_thickness));
L_c = props.f_height/2 + props.f_thickness/2;
eta_f = tanh(m*L_c)/(m*L_c);
```

```
eta_0 = 1-(A_secondary/(A_primary+A_secondary))*(1-eta_f); %surface efficiency
A_air_eff = eta_0*(A_primary+A_secondary);
```

While fin efficiency measures how well a fin performs against its own ideal, fin effectiveness ϵ_f asks a different question: is the fin worth adding at all, compared to leaving that surface bare? It is defined as the ratio of actual fin heat transfer to the heat transfer that same base footprint would provide with no fin present, $\epsilon_f = \eta_f (A_{\text{secondary}}/A_{\text{c_b}})$, where $A_{\text{c_b}}$ is the total root area the fins occupy on the tube wall. As a rule of thumb, $\epsilon_f > 2$ justifies the added weight and pressure-drop cost of finning a surface; automotive radiator fins typically return values well into the tens, since the thin, tall fin geometry gives a large area ratio even after the efficiency penalty is applied.

```
% fin effectiveness
A_c_b = f_thickness*d*n_tubes*2*fpi*W/0.0254;
epsilon_f = eta_f*(A_secondary/A_c_b);
epsilon_f_mat(i,j) = epsilon_f;
```

FPI	FinEffectiveness
8	76.679
9	76.561
10	76.447
11	76.338
12	76.232
13	76.129
14	76.03
15	75.932
16	75.837
17	75.744
18	75.652
19	75.562
20	75.474
21	75.387
22	75.301
23	75.217
24	75.133
25	75.05

Figure 2: Fin effectiveness values for varying values of fpi

2.9 NTU, ϵ , and Predicted Heat Rejection

This is computed across the full $A_{\text{face}} \times fpi$ grid, producing a 2D map of predicted heat rejection.

```
NTU = UA/C_min; %number of transfer units
epsilon = 1-exp((NTU^0.22/C_star)*(exp(-C_star*NTU^0.78)-1));

Q = epsilon*C_min*(T_coolant_in-T_air_in);
```

2.10 Extracting the Design Point

A filled contour plot of $Q_{\text{predicted}}$ over the $f_{\text{pi}}/A_{\text{face}}$ grid is generated, with the Q_{target} contour and a $Q_{\text{target}} \times 1.1$ (10% safety margin) contour overlaid. This is a design-margin plot: any point above/right of the Q_{target} line is a geometry that clears the heat rejection requirement. The exact minimum frontal area at a chosen f_{pi} is then extracted numerically by pulling the target- Q contour line coordinates (contourc) and interpolating along it (interp1) at the desired f_{pi} , and the required radiator width is derived from A_{face}/H and plotted the same way.

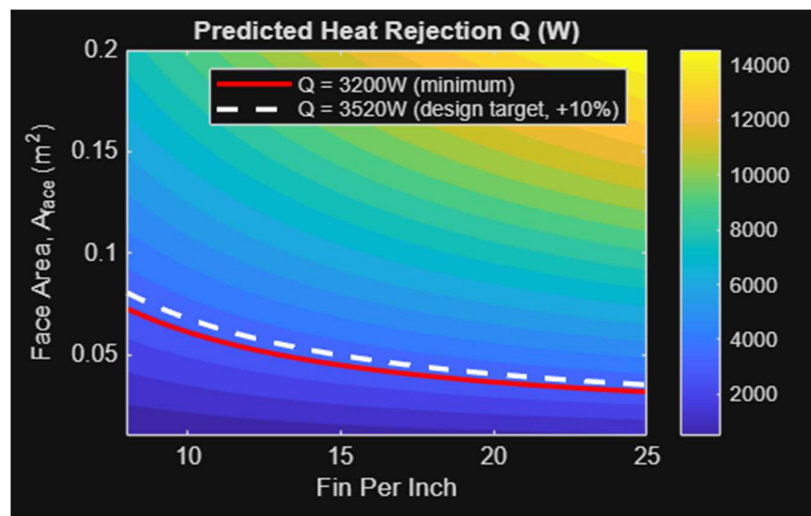


Figure 3: Predicted heat rejection $Q(W)$ over the $f_{\text{pi}} / A_{\text{face}}$ design grid, with the Q_{target} and +10% margin contours overlaid.

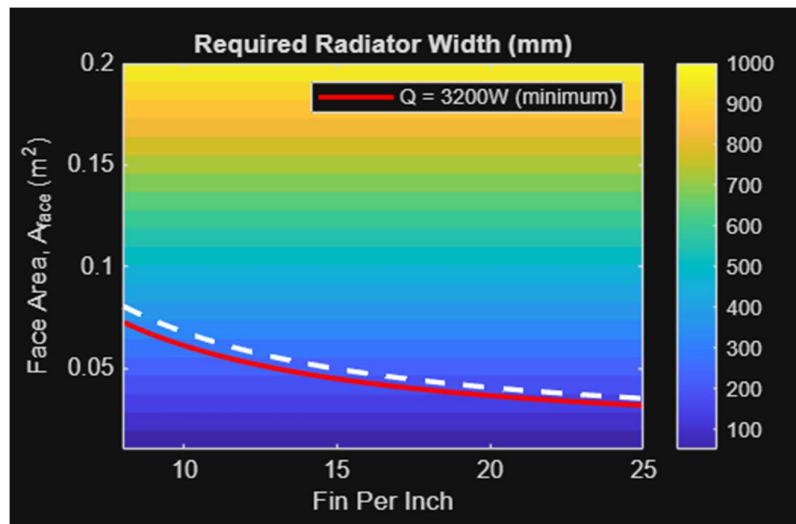


Figure 4: Required radiator width (mm) over the same design grid, with the Q_{target} contour overlaid.

2.11 Water-Side Pressure Drop

At the extracted design point, a Darcy friction factor for laminar flow ($f = 64/Re$) is used to estimate the pressure drop across the radiator core. This is a first-pass check only (a more detailed procedure is planned once pump selection is underway); it covers the water side only, **air-side pressure drop and fan/pumping power (if used) are not modelled**.

For darcy friction factor, again Shah & London curve fit it used to get the factor depending upon aspect ratio and then a correction factor is applied for not fully developed flow.

At $f_{pi} = 18$, minimum A_{face} ($Q=3200W$) = 0.0391 m^2
At chosen f_{pi} and A_{face_min} ($H=200\text{mm}$), Pressure drop = 103585.17 Pa

At $f_{pi} = 18$, design target A_{face} ($Q=3520W$, $+10\%$) = 0.0434 m^2
At chosen f_{pi} and A_{face_min} ($H=200\text{mm}$), Pressure drop = 113069.75 Pa

Flag: this is only the pressure drop across the radiator core itself. It does not account for the rest of the cooling circuit (motor, motor controller, pump, and connecting pipes), so it should not be read as the total system pressure drop the pump needs to overcome.

2.12 Sensitivity Studies

Two one-way sensitivity sweeps are run, holding the chosen design point's f_{pi} and A_{face} fixed:

- **Height (H) sweep, 160–250 mm:** shows that at fixed f_{pi} and frontal area, varying height has a negligible effect on predicted Q . This justifies treating H as a fixed 200 mm parameter in the main sweep rather than a third swept dimension. (The zig-zag pattern is due to the number of tubes in the radiator being a step function and while height increases for a fixed area, width decreases, which further reduces both $A_{primary}$ and $A_{secondary}$.)

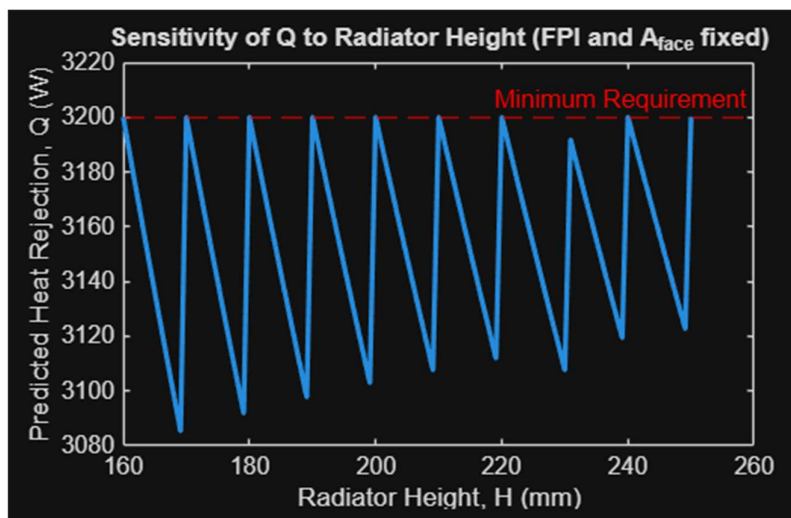


Figure 5: Sensitivity of predicted Q to radiator height H , with f_{pi} and A_{face} fixed at the design point.

- **Coolant flow rate sweep, 3 - 21L/min:** shows Q rising quickly at low flow rate (where $C_{min} = C_{water}$, i.e. the coolant side is the bottleneck) before flattening out once flow rate crosses the point where C_{min} switches to the air side ($C_{min} = C_{air}$). Past that crossover, more coolant flow barely changes Q because heat rejection is now bottlenecked by the air side. This crossover flow rate is calculated explicitly and marked on the plot alongside the current pump's rated flow.

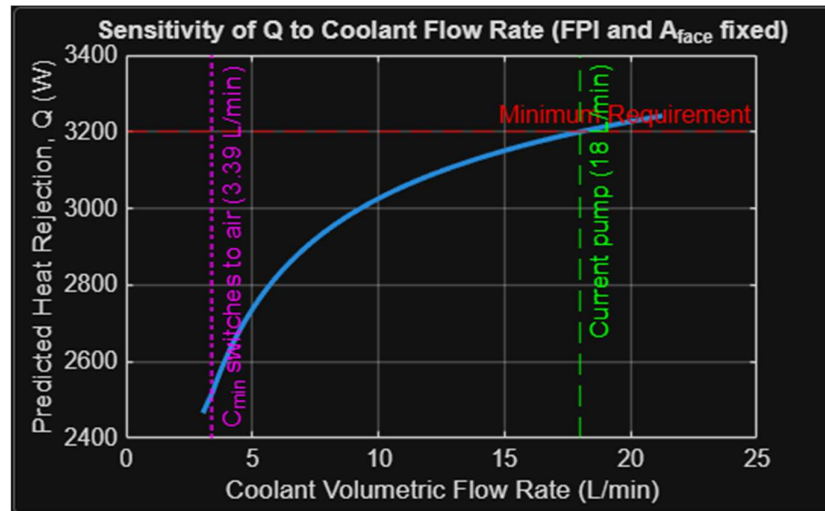


Figure 6: Sensitivity of predicted Q to coolant mass flow rate, showing the C_{min} crossover from water-side- to air-side-limited.

2.13 A_{face} vs. FPI Summary Table

A final summary loop prints the minimum required frontal area (and corresponding width) at each swept fpi value, giving a quick reference table for trading off fin density against radiator size.

FPI	A_{face} (m ²)	Width (mm)
8	0.0727	363.5
9	0.0663	331.5
10	0.0610	305.2
11	0.0566	283.2
12	0.0529	264.7
13	0.0498	248.9
14	0.0470	235.2
15	0.0447	223.4
16	0.0426	213.0
17	0.0408	203.9

FPI	A_face (m ²)	Width (mm)
18	0.0391	195.7
19	0.0377	188.5
20	0.0364	182.0
21	0.0352	176.2
22	0.0342	171.0
23	0.0332	166.2
24	0.0324	161.9
25	0.0316	157.9

3. Transient Models (radiator_transient_fmot.mlx / radiator_transient_emrax.mlx)

The steady-state script sizes the radiator against a single conservative worst-case operating point. The transient model instead replays a real endurance-run data set through the chosen geometry and sweeps the coolant temperature with time, to check whether the design survives the actual, time-varying accel/brake/idle profile of an event rather than just the theoretical worst case. radiator_transient_fmot.mlx and radiator_transient_emrax.mlx are structurally identical; the only difference between them is the motor + controller efficiency map, since each motor has its own datasheet efficiency contours and different max input values of torque and power possible.

3.1 Extracting Data as Input

Data is read from a logged endurance run (FSPT_endurance.xlsx, a placeholder filename for whatever file is being used to record data):

```
clear; clc; close all;
dataTable = readtable('FSPT_endurance.xlsx'); %placeholder name
t_data = dataTable.Timestamp/1000000; %(s)
torque_data_excel = dataTable.Torque_Right;
torque_data = torque_data_excel*(67.88/32767); %(Nm)
tsv = dataTable.TSV; % Tractive System Current
tsc = dataTable.TSC; % Tractive System Voltage
power_data = tsv .* tsc; %(W)
rpm_data = dataTable.RPM_RR;
```

3.2 Data Cleaning: Zero-RPM Filtering and Time Shifting

Samples where the motor is not spinning are dropped, and the remaining timestamps are re-accumulated so the simulation timeline is continuous (no gaps left behind by the removed samples). This keeps the model focused on periods where the motor is actually generating heat and moving the car.

```
valid_idx = rpm_data ~= 0;
dt_orig = [0; diff(t_data)];
t_data = cumsum(dt_orig(valid_idx));
torque_data = torque_data(valid_idx);
power_data = power_data(valid_idx);
rpm_data = rpm_data(valid_idx);
```

Flag: this compresses out real elapsed time during idle/stationary periods. Since $Q_{\text{gen}} \approx 0$ at zero RPM this is a reasonable approximation for heat generation, but it does mean the simulated timeline no longer matches the real event duration, any heat soak or radiator cooling that would have continued to occur during idle (e.g. between endurance stints) is not represented.

3.3 Motor + Controller Efficiency Map

RPM–torque–efficiency data points from the motor + controller combined efficiency map (datasheet contour lines, hand-extracted into `rpm_pts/torque_pts/eff_pts` triplets) are interpolated with a scattered interpolant:

```
F = scatteredInterpolant(rpm_pts(:), torque_pts(:), eff_pts(:), 'linear',
    'nearest');
```

At each timestep, efficiency is looked up at the current (RPM, torque) operating point, clamped to [0,1], and used to convert electrical input power into dissipated heat:

```
% --- heat generated ---
eff = F(rpm_data(k), torque_data(k));
P_check = torque_data(k)*rpm_data(k)*2*pi/60; % (W)
if torque_data(k) > 31.5 || power_data(k) > 35000
    warning('t=%.2fs: T=%.1f Nm, RPM=%.0f, Logged P=%.0f W, computed P=%.0f W',
        ...
            t_data(k), torque_data(k), rpm_data(k), power_data(k), P_check);
end
eff = min(max(eff, 0), 1); % efficiency between 0-1 (decimal)
Q_gen = power_data(k)*(1-eff); %both motor and motor controller
Q_gen_history(k) = Q_gen;
```

This is a **combined** motor+controller loss figure straight from the map, in contrast to the steady script's separate motor-loss + fixed-750 W-MC accounting.

Flag: lines 192–194 above are warning messages added when the data was outputting more torque or power than the motor can produce. The current FSPT_endurance.xlsx data is from the older motors, which have a higher limit than the newer motors - running this code with that data will spam warning messages until it is replaced with F-MOT telemetry.

3.4 Initial Parameters (Flagged for Iteration)

The geometry values are meant to be carried over from whatever design point was extracted in the steady script. gear_ratio and ram_air_factor are marked as placeholder values, pending whatever values are decided by the drivetrain/aero team in the future.

```
% MAKE CHANGES HERE
A_face_fixed = 0.0286;
fpi_fixed = 18;
H_fixed = 0.35; %(m)
d = 0.032; %(m), depth of radiator
W = A_face_fixed/H_fixed; %(m), width of radiator
mdot_coolant = 10; %(L/min), per side, fixed before the loop
T_air_in = 25; %(degC)
T_coolant_initial = 25; %(degC)
r_wheel = 0.11; %(m)
gear_ratio = 4.5;
ram_air_factor = 0.7; %placeholder value
```

3.5 Deriving Air Velocity from Wheel Kinematics

Rather than assuming a single fixed v_{air} like the steady script, the transient model derives it from the car's actual speed at each timestep, via motor RPM and the drivetrain ratio:

```
% --- vehicle & air speed ---
omega_motor = rpm_data(k)*2*pi/60; % (rad/s)
omega_wheel = omega_motor/gear_ratio;
v_car = r_wheel*omega_wheel; % (m/s)
v_air = ram_air_factor*v_car; % (m/s)
v_air = max(v_air, 0.5); % floor to avoid v_air = 0 (stalled/idle) blowing up
h_air
v_air_history(k) = v_air;
```

Flag: ram_air_factor depends on the placement of the radiator on the car, which is to be decided by the aero team, but a position can't be decided unless we provide them with the required air mass flow rate through the radiator, which itself depends on v_{air} into the radiator. Future discussion required on this issue.

3.6 Q_epsilonNTU Function

The steady-state script's inline heat-rejection calculation is refactored into a standalone function so it can be called once per timestep:

```
function [Q_rejected, water_state] = Q_epsilonNTU(T_coolant_in, v_air,
mdot_coolant_kgs, A_face, fpi, H, props, water_state)
```

Two performance considerations are built in:

- **props struct:** all the fixed geometry and air-property constants are packed into a struct once, outside the time loop, instead of being recomputed every timestep.
- **water_state:** water properties (k , μ , Pr) are only re-queried from CoolProp when the coolant temperature has moved by $\geq 1^\circ\text{C}$ since the last lookup, rather than on every single timestep (to reduce computational time for long python calls required by CoolProp). Since these properties change slowly with temperature, this significantly cuts down on repeated CoolProp/Python calls over long-running data without changing accuracy significantly. **Air properties, by contrast, remain fixed at the 25°C reference value for the entire run.**

3.7 Time-Marching Loop (Lumped Capacitance, Explicit Euler)

The coolant in the radiator core is treated as a single lumped thermal mass:

```
m_water = props.rho_water*props.A_tube*W*n_tubes; % mass of water held in the
radiator core
```

Flag: this is the mass of coolant in the radiator core only, not the full coolant loop volume (motor jacket, motor controller, pipes, pump). The lumped-capacitance model is only as good as this thermal mass estimate; using only the radiator's own volume likely understates the total thermal mass of the loop, which would make the simulated temperature respond faster (both heating and cooling) than the real system.

Each step is a first-order forward (explicit) Euler update:

```
T_new = T_old + dt*(Q_gen - Q_rejected)/(m_water*cp_water);
```

The full temperature history is stored for every timestep (not just the final value), so the whole thermal history can be inspected. Timesteps where the 70°C limit is exceeded are logged (first-exceedance time, peak temperature) without breaking the loop, so the full duration of any exceedance is captured rather than just its first occurrence.

The 70°C upper limit is chosen as the closest-to-safe operating temperature of the coolant that the motor controller (being a highly temperature-sensitive device) can feasibly operate on.

3.8 Outputs

- Coolant temperature vs. time, with a 70°C limit reference line.

- Peak temperature and time of first limit violation (or confirmation the run stayed under limit).
- Q_{gen} vs. Q_{rejected} over time: visualizes where in the run the radiator is keeping up vs. falling behind.
- A post-warm-up minimum temperature check (looking only after a threshold time, e.g. 100 s in) to see where the system settles once initial transients have died out.

4. F-MOT vs. EMRAX: What's Different

The two transient scripts are near-identical: same data pipeline, same $Q_{\text{epsilonNTU}}$ function, same time loop, same outputs. The **only** difference between them is the efficiency map ($\text{rpm_pts}/\text{torque_pts}/\text{eff_pts}$) used to build the `scatteredInterpolant`, since each motor + controller combination has its own datasheet efficiency contours — and the maximum possible torque and power values used for the warning message.

5. Validation Suite (`radiator_transient_test.mlx`)

To verify proper working of the transient model, it was necessary to check that if a constant load (RPM and torque) is given as input, the coolant temperature converges to a value equal to the independently solved steady-state temperature.

5.1 Independent Steady-State Solver

Rather than trusting the time-marching result on its own, each test case is cross-checked against a steady-state coolant temperature found by root-finding, completely independently of the time-stepping loop. This solves $Q_{\text{gen}} = Q_{\text{rejected}}(T_{\text{ss}})$ directly: the temperature at which heat generated exactly balances heat rejected, using bisection between the ambient temperature and 150°C. If the sign of the residual doesn't change across that bracket, no steady-state solution exists in range.

```
% independent steady-state solve:  $R(T) = Q_{\text{gen}} - Q_{\text{rejected}}(T)$ 
% bisection method
residual = @(T) Q_gen-Q_epsilonNTU(T, c.v_air, mdot_coolant_kgs, A_face_fixed,
fpi_fixed, H_fixed, props, []);
T_lo = T_air_in+0.01; % bounds for temp search range, temp cant be below t_air
T_hi = 150; % 150 taken as an upper bound as coolant never reaches this temp
R_lo = residual(T_lo);
R_hi = residual(T_hi);

if Q_gen == 0 % no heat Load means the coolant just relaxes to ambient air temp
    T_ss = T_air_in;
    status = 'converged';
    diff_val = T_final_sim-T_ss;
else
    T_lo = T_air_in+0.01;
    T_hi = 150;
```

```

R_lo = residual(T_lo);
R_hi = residual(T_hi);

if sign(R_lo) == sign(R_hi)
    T_ss = NaN;
    status = 'NO STEADY SOLUTION (overload) - check T_final_sim is
monotonically rising';
    diff_val = NaN;
else
    T_ss = fzero(residual, [T_lo T_hi]);
    status = 'converged';
    diff_val = T_final_sim-T_ss;
end
end

```

5.2 Test Case Matrix

Seven constant-condition cases (held RPM/torque/air velocity, run out to a fixed duration) probe different aspects of the model's correctness:

Case	Conditions	What it checks
A: Zero load	RPM=5000, torque=0	With no heat generated, coolant temperature should relax to ambient air temperature.
B: Nominal design point	RPM=8000, torque=20 Nm, $v_{air}=10$	A representative mid-map operating point, approached from a cold start ($T_{initial} = 25^{\circ}\text{C}$).
C: Overload / stall	RPM=2000, torque=26 Nm (near max), v_{air} floored at 0.5	Heat generated exceeds the maximum heat the radiator can reject at that air velocity, no steady-state solution exists (in the 70°C limit). Confirms the model correctly identifies a genuine thermal runaway condition rather than silently producing a wrong equilibrium.
D: Nominal, approached from above	Same as B, but $T_{initial} = 80^{\circ}\text{C}$	Confirms the same equilibrium temperature is reached whether approached from below (case B) or above (case D).
E1 / E2 / E3: Air velocity sweep	Same nominal Q_{gen} as B, $v_{air} = 5, 15, 30$ respectively	Confirms monotonic behaviour: higher air velocity $\rightarrow$ more heat rejected $\rightarrow$ lower equilibrium coolant temperature (E1 > E2 > E3 in final temperature).

5.3 Time Constant and t_{99}

For each case, a thermal time constant is estimated from the final-state conductance:

```
tau_est = m_water*cp_water/UA_final;
```

and t_{99} : the time to close 99% of the gap between the initial and final (independently solved) steady-state temperature, found by linear interpolation between the two simulated samples that bracket the 99% point. For a first-order lumped system, $t_{99} \approx 4.6 \cdot \tau_{est}$ (since $\ln(100) \approx 4.6$); this relationship is checked directly in the printed summary as a sanity check that the simulation is reproducing expected first-order step-response behaviour when $dt \ll \tau_{est}$.

5.4 Summary Table and Pass/Fail Checks

The script prints a table of Q_{gen} , simulated final temperature, independently-solved steady-state temperature, their difference, τ_{est} , t_{99} , and status for all seven cases, followed by explicit written checks to manually verify against:

- $|diff|$ should be small (sub-0.1°C) for cases A, B, D, E1–E3 when $dt \ll \tau_{est}$.
- Case C should report converging temperature higher than feasible operating range.
- $E1 > E2 > E3$ in final temperature.
- Case D should converge to the same steady-state temperature as case B.
- $t_{99} \approx 4.6 \cdot \tau_{est}$ across the cases with a defined steady state.

Only once these checks pass is the same $Q_{epsilonNTU}/\text{time-sweep}$ combination considered trustworthy enough to run against real data in the F-MOT/EMRAX scripts.

6. COMSOL Validation of the Steady-State Model

The steady-state design script (Section 2) predicts heat rejection from general air-side and water-side convection correlations combined with the ϵ -NTU method. To check that this prediction is physically reasonable, a conjugate heat transfer model of a single radiator tube was built in COMSOL Multiphysics, and the heat rejected per tube, Q , was compared against the value predicted by `radiator_steady_final.mlx` for the same geometry and operating conditions. As all tubes in the core are nominally identical, the heat rejected per tube was chosen as the comparison metric.

Reaching a converged and usable model took eleven iterations. The early iterations resolved both the air and water flows in a scaled-down radiator and failed on mesh quality and solver convergence. The final model replaces the air flow with a boundary condition based on the MATLAB model, which made the problem tractable. The iterations are summarized below and discussed in the following subsections.

Iteration	Main change	Outcome
1	Scaled-down radiator (4 tubes, 3 fin channels), default mesh, boundary layer on air domain only	Minimum element quality 0.01–0.05; did not converge
2	λ continuation parameter on flow rates, pseudo time stepping on, swept mesh for the fins	No λ converged; minimum element quality ≈ 0.03
3	Water inlet changed to volumetric flow rate	Did not converge; abandoned
4	Auxiliary sweep with continuation, small starting λ	Converged up to $\lambda = 0.04$ in ≈ 5.5 h; abandoned

Iteration	Main change	Outcome
5	Turbulent flow model with Iteration 4 settings	Converged worse than laminar model
6	Reduced to 1 tube, 2 fin channels; fins modelled as one folded sheet	Swept mesh no longer possible; element quality dropped further
7	Boundary layers across fin thickness and on both fluid domains	Mesh build abandoned; tube wall found to be missing from model
8	Periodic unit cell (two half-tube channels and one fin channel)	Geometry set up; not run
9	Air domain replaced by convective boundary condition at constant ambient air temperature; tube wall included; fillets removed	Converged; extrapolation of results to full channel length unreliable
10	Full-length water channel simulated	$Q_{\text{air}} = 287.78$ W per tube; large deviation from MATLAB
11	Air temperature solved as an energy-balance equation along the air path	$Q_{\text{air}} = 217.08$ W per tube; 8.88% above MATLAB

Flag: since h_{air} in the COMSOL model is taken from the MATLAB script, this comparison does not independently validate the air-side convection coefficient (the Colburn j-factor flagged in Section 2.6). It validates the remainder of the chain: water-side convection, conduction through the tube and fins, and the ϵ -NTU formulation.

6.1 Full Air and Water Domain Simulations (Iterations 1–5)

The first iterations modelled both fluid domains explicitly. Iteration 1 used a scaled-down radiator of four tubes and three fin channels, with mostly default mesh settings and a boundary layer applied only to the air domain (Figure 7). The study did not converge. The principal problem in almost all of the early iterations was mesh quality: the minimum element quality was generally between 0.01 and 0.05, well below the value of 0.1 normally required for CFD. The very small fin thickness relative to the other dimensions also triggered a mesher warning (Figure 8).

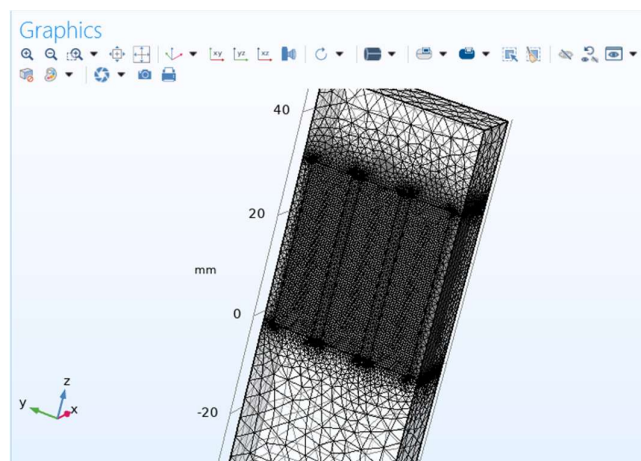


Figure 7: Mesh of the scaled-down radiator in Iteration 1.

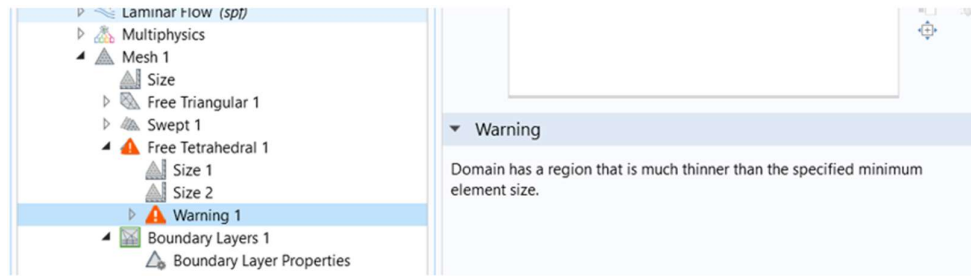


Figure 8: Mesher warning for the thin fin regions in Iteration 1.

In Iteration 2, a continuation parameter λ (λ), ranging from 0 to 1, was introduced and used to scale the water mass flow rate and the air velocity. The intention was that a solution obtained at a small λ would converge more easily and could then be used as the initial guess for the next, larger λ , progressively reaching the full operating condition. No value of λ converged. Pseudo time stepping was also changed from automatic to on, although its effect could not be established. A swept mesh was added to deal with the small fin thickness; this removed the mesher warnings, but the mesh still contained an excessive number of elements (approximately 1.77 million) and the minimum element quality remained low at about 0.03 (Figure 9).

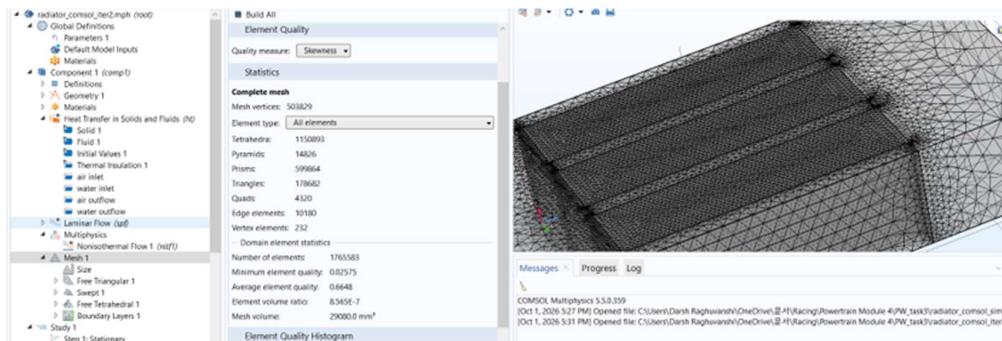


Figure 9: Mesh statistics after the swept mesh was added in Iteration 2.

In Iteration 3 the water inlet was specified as a volumetric flow rate, obtained by dividing the mass flow rate by the water density and scaling with λ (Figure 10), instead of as a mass flow rate. λ was started at 0.1, which did not converge, and the iteration was abandoned. The mesh was not meaningfully changed, so the minimum element quality remained poor.

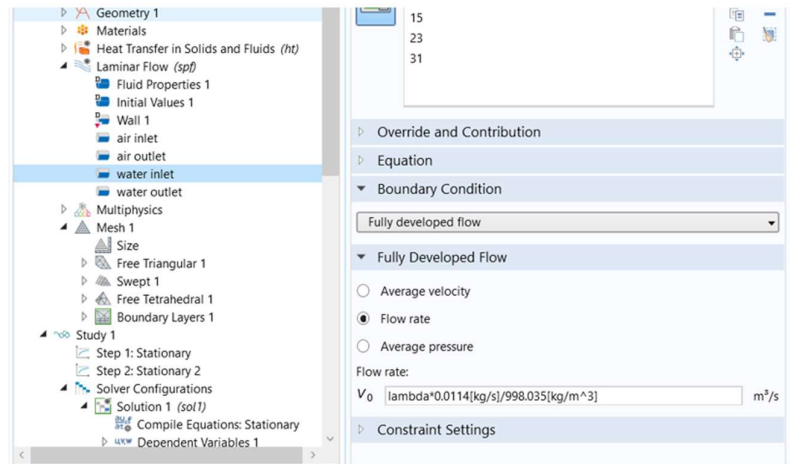


Figure 10: Water inlet boundary condition specified as a fully developed flow with a flow rate scaled by λ .

Iteration 4 added an auxiliary sweep with continuation, which automatically inserts an intermediate λ value when a step fails (for example, $\lambda = 0.037$ was inserted because the step to 0.04 initially failed), together with much smaller starting values of λ . Starting from $\lambda = 0.01$, the following values converged: 0.01, 0.011, 0.013, 0.017, 0.022, 0.03, 0.035, 0.037 and 0.04. Although this showed more progress than the earlier iterations, reaching only $\lambda = 0.04$ took approximately 5.5 hours, so the approach was abandoned as impractical. A single run with a turbulent flow model was also carried out in Iteration 5 using the Iteration 4 settings; as expected, it converged even worse than the laminar model.

6.2 Reduced Geometry and Modelling Flaw (Iterations 6–7)

To aid convergence, the model was reduced further in Iteration 6 to one tube with two fin channels (Figure 11). On reviewing the CAD of the earlier radiator, it was recognized that the fins are not separate rectangular plates but a single continuous sheet that is rolled into the fin shape, and the geometry was remodelled accordingly. The folded topology does not support a swept mesh, and the minimum element quality dropped further, to below 0.01 (Figure 12).

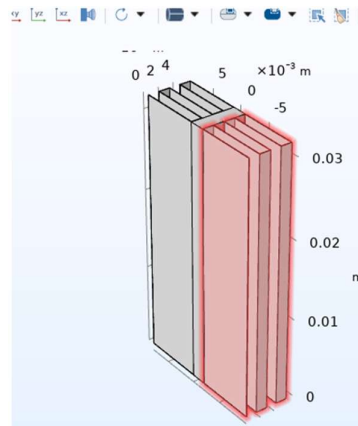


Figure 11: Reduced geometry of one tube and two fin channels (Iteration 6).

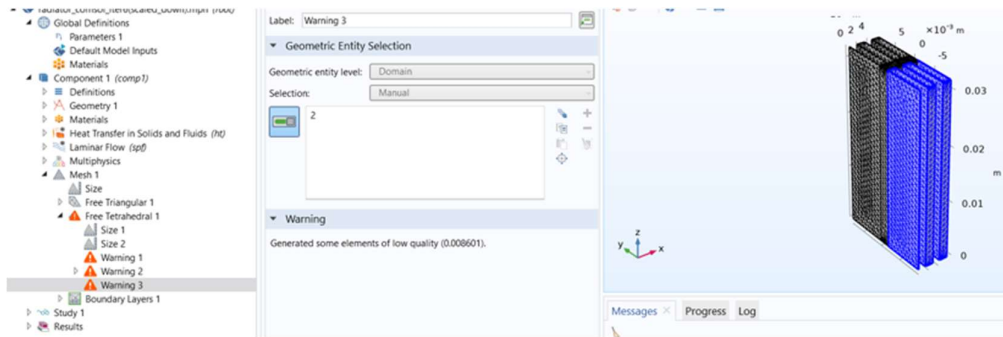


Figure 12: Mesher warning of low-quality elements after the fins were modelled as a folded sheet (Iteration 6).

In Iteration 7 a new meshing approach was tried for the fin thickness: instead of using a very small element size, a boundary layer was placed between the two end faces of the fins (Figure 13). Mesh generation with this setup took very long and was abandoned before completion. While setting up this boundary layer, a more fundamental flaw in the model was identified. Boundary layers had been defined only for the air domain, although the model contains two fluids. When boundary layers were also defined for the water domain, the interface at which the air and water meet had to be selected for both, and a single face cannot carry two boundary layers. This showed that the tube carrying the water was not part of the model: the water and air domains were in direct contact, and a no-slip condition at an interface between two fluids is not physically meaningful.

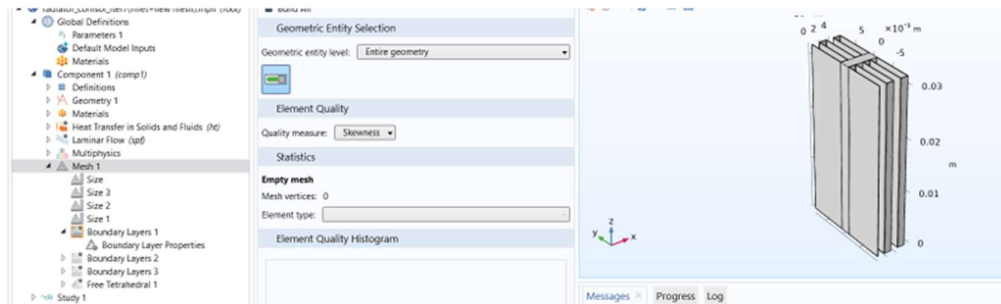


Figure 13: Boundary layer setup across the fin thickness (Iteration 7); the mesh build was abandoned.

The issue originated from the MATLAB model, which neglects the thermal resistance of the tube wall (Section 2.7); the tube was consequently left out of the CFD geometry as well. In a real radiator, heat flows from the water through the tube wall and the fins into the air (water → tube and fin → air), so the tube wall must be modelled explicitly.

6.3 Periodic Unit Cell Model (Iterations 8–10)

After reviewing the literature on CFD of radiators, a periodic unit-cell methodology was adopted (Figure 14). As the core consists of repeating tube and fin layers, a single repeating cell bounded by periodic boundary conditions represents the whole core. The cell consists of two half-tube water channels and one fin channel (Figure 15). This geometry was set up in Iteration 8 but not run, as a further simplification was identified. The tube wall was introduced in Iteration 9.

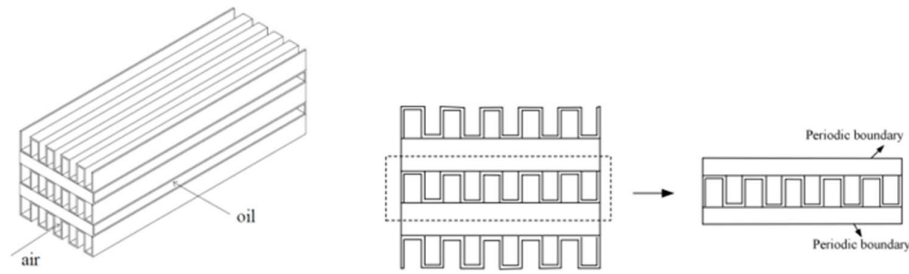


Fig.1 The periodic structure of exchanger

Figure 14: Periodic structure of a heat exchanger core and the resulting unit-cell reduction, reproduced from the literature on which the approach is based.

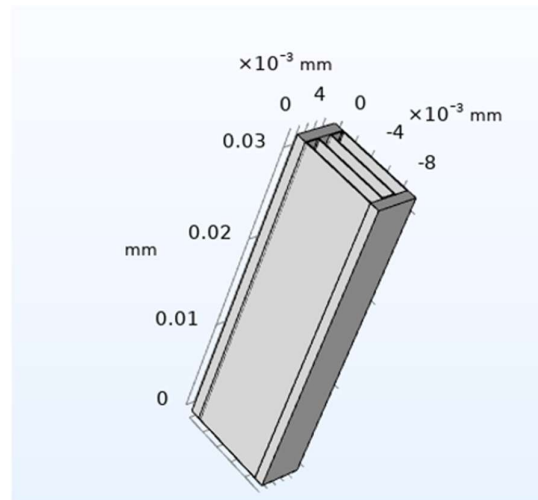


Figure 15: CAD of the periodic unit cell.

As the vehicle is moving, fresh air flows continuously through the radiator, so the air temperature was initially assumed to remain constant at its ambient value. Rather than simulating the air domain, Iteration 9 therefore applied the convective heat transfer coefficient h_{air} computed by the MATLAB script as a boundary condition on the outer surface of the tube and on the fin faces in contact with air. The fillets were also removed, as they had been included for manufacturing reasons and do not significantly affect the fluid flow so their inclusion only made the CFD convergence harder. With these changes the model converged (Figure 16).

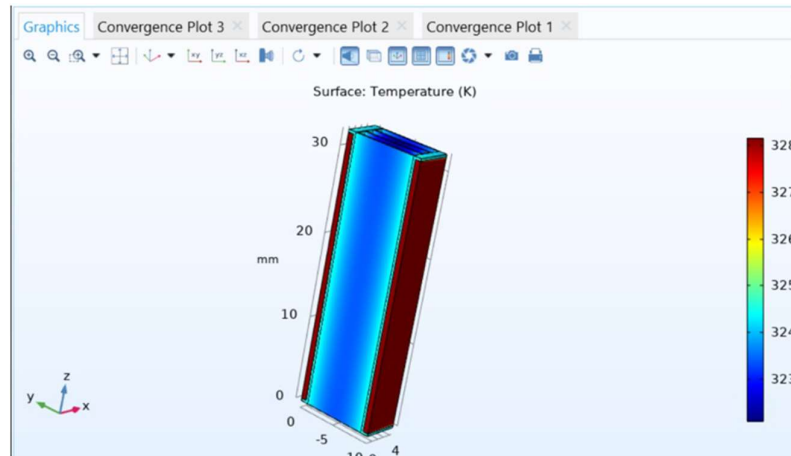


Figure 16: Temperature field of the unit cell with a constant air temperature boundary condition (Iteration 9).

Extrapolating the temperature data from this model to the full channel length was attempted using a polynomial fit and an exponential fit. The polynomial fit gave extremely poor values, while the exponential fit converged to a value that differed drastically from the MATLAB prediction. An entire water channel was therefore simulated in Iteration 10 (Figure 17).

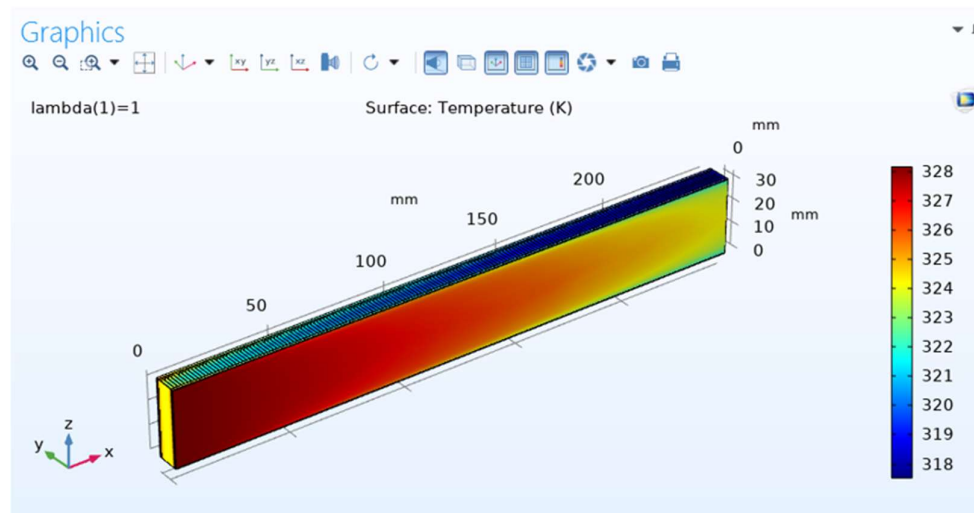


Figure 17: Temperature field over the full water channel length (Iteration 10, $\lambda = 1$).

This simulation gave $Q_{air} = 287.78$ W per tube, a large deviation from the MATLAB estimate at that stage of approximately 160 W. The MATLAB calculation was found to contain an error and was reworked, after which it predicted approximately 198 W per tube. Most of the remaining difference with COMSOL was attributed to the assumption of a constant air temperature on the tube and fin surfaces, which overestimates the temperature difference driving heat transfer to the air, since in reality the air warms as it passes through the core.

6.4 Air Temperature as a Boundary Condition (Iteration 11)

To remove this error without simulating the air domain, the local air temperature T_a was introduced as an additional dependent variable, governed by a standard energy balance on the air stream and coupled to the convective boundary condition on the tube and fin surfaces:

$$(\dot{m}_{\text{cell}} \cdot c_p / W_{\text{perp}}) \cdot dT_a / ds = h \cdot (T - T_a)$$

The first term is the heat capacity rate of the air flowing through the cell per unit width of air channel (W_{perp}), s is the coordinate along the air flow direction, h is the air-side convection coefficient, T is the local wall temperature and T_a is the local air temperature. The air therefore warms along its path through the core, which reduces the local driving temperature difference, and the surface heat flux is evaluated against T_a instead of the fixed inlet air temperature (Figure 18).

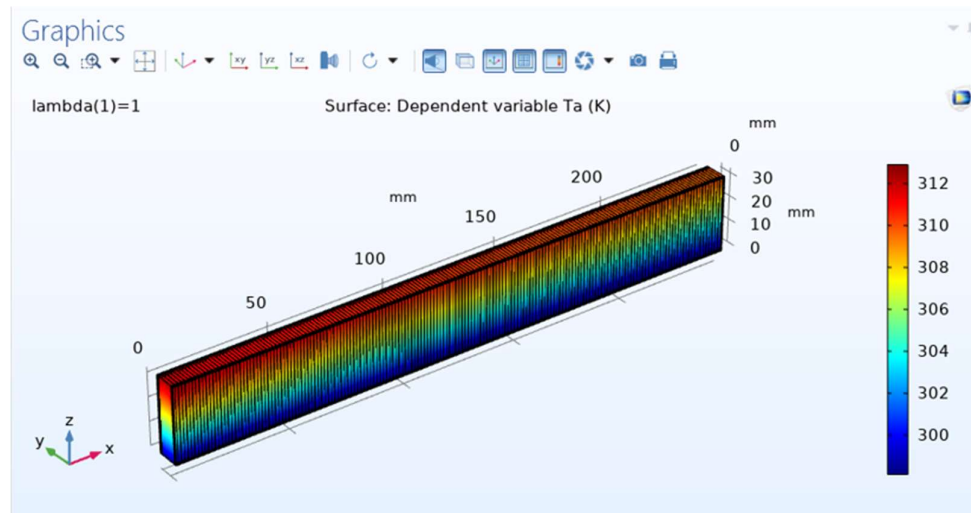


Figure 18: Air temperature T_a over the unit cell (Iteration 11, $\lambda = 1$).

6.5 Implementation Details

In order to setup the h_{air} in comsol, rather than running the radiator_Q function at a certain area $A_{\text{face_min}}$, area was recalculated using W as taken in cad and that number was used to validate the MATLAB script. Q_{tube} & T_{out} were calculated as the major factors for validation. According to iteration 10, the condition of constant air temperature was replicated in MATLAB, with the assumption of C_{air} being infinity which utilises a simplified ϵ -NTU approach to calculate Q . Although this was scrapped as I moved onto iteration 11.



```
% h_air at design point for comsol validation
% (h_air does not depend on A_face, A_face_min is only there to complete the
function call)
[~, info_design] = radiator_Q(A_face_min, fpi_query, H, mdot_coolant_kgs, params);
fprintf('h_air for boundary condition = %.4f', info_design.h_air);

% Q_rejected at a certain Width
W = 0.24685; % cad is 246.85mm in width
fpi = 18;
A_face = W*H;
[Q, info_comsol] = radiator_Q(A_face, fpi, H, mdot_coolant_kgs, params);
Q_tube = Q/info_comsol.n_tubes;
dT_water = Q/(mdot_coolant_kgs*cp_water);
T_out = T_coolant_in - dT_water;
fprintf('Q_rejected_total = %.2fW, Q_rejected_tube = %.2fW', Q, Q_tube);
fprintf('T_out = %.2f degC (%.2f K), dT = %.2f K\n', T_out, T_out+273.15,
dT_water);

% comsol assumes constant temp for air (25degC), we can replicate that in
% matlab with C_air = infinity and using simplified epsilon relation
Q_tube_comsol = 287; % Q_rejected_tube from comsol
UA_tube = info_comsol.UA/info_comsol.n_tubes;
C_water_tube = info_comsol.C_water/info_comsol.n_tubes;
Q_tube_inf_air = C_water_tube*(T_coolant_in-T_air_in)*(1-exp(-
UA_tube/C_water_tube)); % simplified e_ntu, Q = C_min*deltaT*(1-e^(-NTU)) where
NTU = UA/C_min
UA_tube_comsol = -C_water_tube*log(1 - Q_tube_comsol/(C_water_tube*(T_coolant_in-
T_air_in))); % effective UA implied by comsol
fprintf('Q_tube (infinite air) = %.2fW, UA_tube_matlab = %.3f, UA_tube_comsol =
%.3f W/K', Q_tube_inf_air, UA_tube, UA_tube_comsol);

% Air-side inputs for the COMSOL boundary PDE (cb)
n_tubes = floor(H/tube_pitch);
A_face = W*H;
mdot_air_total = rho_air*v_air*A_face; % total air mass flow (kg/s)
mdot_air_cell = mdot_air_total/n_tubes; % air through one tube pitch (kg/s)

W_perp = (info_comsol.A_primary +
info_comsol.A_secondary)/(d*info_comsol.n_tubes); % width of wetted perimeter
(including both tube + fin)
beta = mdot_air_cell*cp_air/W_perp; % W/(m*K), magnitude of beta

fprintf('cp_air = %.2f[J/(kg*K)]\n', cp_air);
fprintf('mdot_air_cell = %.6f[kg/s]\n', mdot_air_cell);
fprintf('W_perp = %.5f[m]\n', W_perp);
fprintf('beta = %.4f[W/(m*K)]\n', beta);
```

For setting up the differential equation in comsol several factors needed to be calculated in MATLAB to be given as inputs in COMSOL. This includes `cp_air`, `mdot_air_cell`, `W_prep` and `beta`.

6.6 Results and Comparison with the MATLAB Model

The final COMSOL model (Iteration 11) was run with a water inlet temperature of 55°C and a water mass flow rate of 0.014616 kg/s per tube. The heat rejected to the air was $Q_{\text{air}} = 217.08$ W per tube, with a water outlet temperature of 324.38 K (51.23°C). As a consistency check, the heat lost by the water was also computed from a water-side energy balance, $Q_{\text{water}} = 0.014616 \times 4181 \times (328 - 324.38) = 221.22$ W, which is 1.9% higher than Q_{air} .

This energy balance is very sensitive to the outlet temperature: a change of only 0.1 K in the outlet temperature (324.48 K) gives $Q_{\text{water}} = 215.1$ W, a shift of approximately 6 W. The difference of approximately 4 W between Q_{air} and Q_{water} therefore lies within the resolution of the reported outlet temperature, and the COMSOL solution is considered consistent with itself.

Flag: the inlet water temperature was specified as 55°C (328.15 K), but COMSOL reports it as 328 K in its output. Taking 328.15 K in the water-side balance gives $Q_{\text{water}} = 230.38$ W, which is 6.12% above Q_{air} . The internal discrepancy of the COMSOL solution is therefore approximately 2–6%, depending on the inlet temperature taken. The origin of the 328 K value is yet to be confirmed.

The MATLAB prediction for the same tube is compared with the COMSOL result below.

Quantity	COMSOL	MATLAB	Difference
Heat rejected per tube, Q (W)	217.08	197.79	–8.88% (MATLAB lower)
Water outlet temperature (°C)	51.23	51.81	+0.58 °C (MATLAB higher)

The MATLAB model under-predicts the heat rejected per tube by 8.88% relative to COMSOL and over-predicts the water outlet temperature by 0.58 K. This deviation includes the 2–6% internal discrepancy of the COMSOL solution itself, so it cannot be attributed entirely to the MATLAB model. As the MATLAB model under-predicts heat rejection, the deviation is on the conservative side for radiator sizing.

7. Known Assumptions and Simplifications

Consolidated from all four scripts, roughly in order of how likely they are to matter:

- **Thermal mass scope:** m_{water} in the transient models is the coolant mass held in the radiator core only, not the full coolant loop (motor, controller, hoses, pump). Likely understates total system thermal mass.
- **Ram-air factor:** $\text{ram_air_factor} = 0.7$ is a placeholder constant relating car speed to actual air velocity through the core; not yet derived from aero/CFD data.
- **Colburn j-factor correlation:** $J = 0.174/\text{Re}^{0.383}$ is inherited from the prior code base; its source and applicability to this specific fin geometry has not yet been independently confirmed against Kays & London reference data.



- **Motor controller loss accounting:** the steady script uses a fixed 750 W MC dissipation figure on top of motor loss; the transient scripts pull a combined motor+MC loss straight from the efficiency map.
- **Air-side properties fixed:** air density, viscosity, conductivity, and Prandtl number are all evaluated once at 25°C and held constant for the entire run; only water properties are updated as temperature changes.
- **Air-side pressure drop / fan power:** not modelled, as fan usage for the radiator is not yet decided. Only water-side pressure drop (laminar Darcy friction factor) is estimated, and only at the design point in the steady script.
- **Fin/tube bond resistance:** assumed negligible, pending manufacturing/bonding process confirmation.
- **Zero-RPM filtering:** dropping and time-compressing idle samples is reasonable for heat generation (which is ≈ 0 at idle), but means the simulated timeline doesn't match real elapsed event time, and doesn't explicitly model any cooling/heat soak that would occur during real idle periods.
- **Single-event scope:** all validation so far is against a single endurance run or synthetic constant-condition case. Thermal ratcheting across multiple repeated endurance stints (i.e. whether the coolant fully recovers between stints, or slowly climbs run-over-run) is not yet modelled or tested.

8. Suggested Workflow

1. Run **radiator_steady_final.mlx** to sweep FPI/frontal area and extract a minimum viable geometry (A_{face} , f_{pi}) against the fixed target heat load, at a fixed worst-case operating point.
2. Feed that geometry into the **% MAKE CHANGES HERE** block of the transient scripts.
3. Run **radiator_transient_test.mlx** first, on synthetic constant-condition cases, to confirm the time-marching implementation numerically agrees with an independently solved steady-state temperature before trusting it on real data.
4. Run **radiator_transient_fmot.mlx** / **radiator_transient_emrax.mlx** against real endurance-run telemetry to check whether the chosen geometry keeps coolant temperature under the 70°C limit across an entire representative run, not just at one static worst-case point.